

STUDENT ATTENDANCE MANAGEMENT SYSTEM

AI23431- WEBTECHNOLOGY AND MOBILE

APPLICATION

MINI-PROJECT REPORT

SUBMITTED BY

PRITIKA.R(241801216)

in partial fulfillment for the award of the degree of

BACHELOR OF TECHNOLOGY

in

ARTIFICIAL INTELLIGENCE AND DATA SCIENCE



**DEPARTMENT OF ARTIFICIAL INTELLIGENCE
AND DATA SCIENCE**

**RAJALAKSHMI ENGINEERING COLLEGE
(AUTONOMOUS),
CHENNAI-602 105**

MARCH 2025

RAJALAKSHMI ENGINEERING COLLEGE

(an Autonomous Institution Affiliated to Anna University, Chennai)

BONAFIDE CERTIFICATE

Certified that this Project Thesis titled “**Online Vehicle Service Booking System**” is the Bonafide work of PRITIKA.R(2116241801216) who carried out the work under my supervision. Certified further that to the best of my knowledge the work reported here in does not form part of any other thesis or dissertation on the basis of which a degree or award was conferred on an earlier occasion on this or any other candidate.

SIGNATURE

Dr. J.M. Gnanasekar M.E., Ph.D.,
Professor and Head
Department of AI&DS
Rajalakshmi Engineering College
Chennai – 602 105

SIGNATURE

MR.G.THIYAGARAJAN
Assitant professor
Department of AI&DS
Rajalakshmi Engineering College
Chennai-602 105

Submitted to Project Viva-Voce Examination held on _____

INTERNAL EXAMINER

EXTERNAL EXAMINER

ACKNOWLEDGEMENT

Initially we thank the Almighty for being with us through every walk of our life and showering his blessings through the endeavor to put forth this report. Our sincere thanks to our Chairman **Mr. S.MEGANATHAN, B.E, F.I.E.**, our Vice Chairman **Mr. ABHAY SHANKAR MEGANATHAN, B.E., M.S.**, and our respected Chairperson **Dr. (Mrs.) THANGAM MEGANATHAN, Ph.D.**, for providing us with the requisite infrastructure and sincere endeavoring in educating us in their premier institution.

Our sincere thanks to **Dr. S.N. MURUGESAN, M.E., Ph.D.**, our beloved Principal for his kind support and facilities provided to complete our work in time. We express our sincere thanks to **Dr. Dr. J.M. Gnanasekar, M.E Ph.D.**, Professor and Head of the Department of Artificial Intelligence and Machine Learning for his guidance and encouragement throughout the project work. We convey our sincere and deepest gratitude to our internal guide, **Mr.G.THIYAGARAJAN**, Department of Artificial Intelligence and Machine Learning for his useful tips during our review to build our project.

PRITIKA.R(241801216)

DEPARTMENT VISION

To promote highly Ethical and Innovative Computer Professionals through excellence in teaching, training and research.

DEPARTMENT MISSION

- To produce globally competent professionals who are motivated to learn emerging technologies and demonstrate innovation in solving real-world problems.
- To promote research activities among students and faculty members that contribute meaningfully to society.
- To instill strong moral and ethical values in professional practice.

PROGRAMME EDUCATIONAL OBJECTIVES(PEO'S)

PEO 1: To equip students with essential background in computer science, basic electronics and applied mathematics.

PEO 2: To prepare students with fundamental knowledge in programming languages, and tools and enable them to develop applications.

PEO 3: To encourage the research abilities and innovative project development in the field of AI, ML, DL, networking, security, web development, Data Science and also emerging technologies for the cause of social benefit.

PEO 4: To develop professionally ethical individuals enhanced with analytical skills, communication skills and organizing ability to meet industry.

PROGRAMME OUTCOMES (POs)

PO 1: Engineering knowledge: Apply the knowledge of Mathematics, Science, Engineering fundamentals, and an engineering specialization to the solution of complex engineering problems.

PO 2: Problem analysis: Identify, formulate, review research literature, and analyze complex engineering problems reaching substantiated conclusions using first principles of mathematics, natural sciences, and engineering sciences.

PO 3: Design/development of solutions: Design solutions for complex engineering problems and design system components or processes that meet the specified needs with appropriate consideration for the public health and safety, and the cultural, societal, and environmental considerations.

PO 4: Conduct investigations of complex problems: Use research-based knowledge and research methods including design of experiments, analysis and interpretation of data, and synthesis of the information to provide valid conclusions.

PO 5: Modern tool usage: Create, select, and apply appropriate techniques, resources, and modern engineering and IT tools including prediction and modeling to complex engineering activities with an understanding of the limitations.

PO 6: The engineer and society: Apply reasoning informed by the contextual knowledge to assess societal, health, safety, legal and cultural issues and the consequent responsibilities relevant to the professional engineering practice.

PO 7: Environment and sustainability: Understand the impact of the professional engineering solutions in societal and environmental contexts, and demonstrate the knowledge of, and need for sustainable development.

PO 8: Ethics: Apply ethical principles and commit to professional ethics and responsibilities and norms of the engineering practice.

PO 9: Individual and team work: Function effectively as an individual, and as a member or leader in diverse teams, and in multidisciplinary settings.

PO 10: Communication: Communicate effectively on complex engineering activities with the engineering community and with society at large, such as, being able to comprehend and write effective reports and design documentation, make effective presentations, and give and receive clear instructions.

PO 11: Project management and finance: Demonstrate knowledge and understanding of the engineering and management principles and apply these to one's own work, as a member and leader in a team, to manage projects and in multidisciplinary environments.

PO 12: Life-long learning: Recognize the need for, and have the preparation and ability to engage in independent and life-long learning in the broadest context of technological change.

PROGRAM SPECIFIC OUTCOMES (PSOs)

A graduate of the Artificial Intelligence and DATA SCIENCE Program will demonstrate

PSO 1: Foundation Skills: Ability to understand, analyze and develop computer programs in the areas related to algorithms, system software, web design, AI, machine learning, deep learning, data science, and networking

PSO 2: Problem-Solving Skills: Ability to apply mathematical methodologies to solve computational task, model real world problem using appropriate AI and ML algorithms. To understand the standard practices and strategies in project development, using open-ended programming environments to deliver a quality product.

PSO 3: Successful Progression: Ability to apply knowledge in various domains to identify research gaps and to provide solution to new ideas, inculcate passion towards higher studies, creating innovative career paths to be an entrepreneur and evolve as an ethically social responsible AI and DS professional.

COURSE OBJECTIVE

- To provide foundational knowledge and practical skills in creating and structuring web pages using HTML, enabling students to build accessible and well-organized websites.
- To understand and practice Embedded Dynamic Scripting on Client-side Internet Programming.
- To implement Server Side Scripting.
- To facilitate students to understand android Application Design
- To help students to gain a basic understanding of Android application development

COURSE OUTCOME

On completion of course students will be able to,

- CO 2: Upon completing the course, students will be able to design and develop basic web pages with proper structure and semantic elements, ensuring accessibility and functionality.
- CO 3: Design and implement dynamic web page with validation using javascript objects and by applying different event handling mechanism.
- CO 4: Design and implement simple webpage to learn JSP and Servlet.
- CO 5: Design and implement simple Application Design.
- CO 6: Identify various concepts of mobile programming that make it unique from programming for other

CO-PO-PSO Mapping

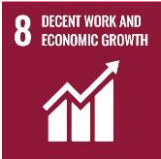
CO	PO 1	P O 2	P O 3	P O 4	P O 5	P O 6	P O 7	P O 8	P O 9	P O 10	P O 11	P O 12	PS O 1	PS O 2	PS O 3
CO 1	3	3	3	3	3	2	2	2	3	2	3	3	3	3	3
CO 2	3	3	3	3	3	2	-	-	2	2	2	3	3	2	2
CO 3	3	3	3	2	3	1	1	2	3	3	3	3	3	3	3
CO 4	3	3	3	3	3	2	1	2	3	2	2	3	3	3	3
CO 5	1	1	1	1	1	-	-	-	3	3	3	3	1	-	2

Note: Correlation levels 1, 2 or 3 are as defined below:

1: Slight (Low) 2: Moderate (Medium) 3:

Substantial (High) No correlation: “-”

SUSTAINABLE DEVELOPMENT GOALS

SDG	GOAL	JUSTIFICATION
 SDG-4	Quality Education	Education drives progress on other SDGs, breaks poverty cycles, and promotes peace and inequality reduction.
 SDG-8	Decent Work and Economic Growth	Necessary for social stability, shared prosperity, and combating poverty.
 SDG-9	Industry, Innovation, and Infrastructure	Infrastructure is vital for economic growth and stability. It acts as a poverty reduction tool through job creation and provides technological solutions to environmental challenges.
 SDG-17	Partnerships for the Goals	Given the interconnected nature of the SDGs, success requires collaborative action, global resource sharing, and bridging the immense financing gap faced by developing nations.

ABSTRACT

The Online Vehicle Service System is a modern web-based application developed to automate and simplify vehicle servicing and maintenance management processes. In traditional vehicle service centers, customers are required to visit workshops physically or contact service advisors manually to schedule service appointments, inquire about service status, and complete billing procedures. These manual methods consume more time, increase paperwork, create communication delays, and often result in inefficient management of customer and vehicle records. The Online Vehicle Service System is designed to overcome these limitations by providing an efficient digital platform for managing all vehicle service-related activities.

The main objective of the system is to provide a secure, reliable, and user-friendly platform that enables customers to book vehicle service appointments online, track service progress in real time, receive notifications regarding maintenance activities, and access complete service history. The system allows customers to register, log in securely, add vehicle information, select service categories, choose preferred appointment dates and times, and make online payments conveniently.

The application also provides separate modules for administrators, service advisors, and mechanics to manage daily workshop operations efficiently. Administrators can monitor customer details, manage appointments, allocate service tasks, maintain spare parts inventory, generate invoices, and analyze workshop performance using centralized management tools. Mechanics and service staff can update service status regularly so that customers receive accurate and real-time updates regarding their vehicles.

In conclusion, the Online Vehicle Service System provides a smart, automated, and efficient solution for modern vehicle servicing operations. The system improves workshop productivity, simplifies customer interaction, enhances service management, and demonstrates how digital transformation can modernize traditional vehicle maintenance systems effectively.

SI NO	TITLE	PAGE NO
	ACKNOWLEDGEMENT	iii
	ABSTRACT	xi
	LIST OF ABBREVIATIONS	xiv
1	INTRODUCTION	1
1.1	General	1
1.2	Need for Online Vehicle Service Booking System	1
1.3	Scope of the System	2
1.4	Organization of the Report	2
1.5	System architecture diagram	3
2	LITERATURE SURVEY	3
2.1	Introduction	3
2.2	Existing Online Vehicle Service Booking System	3
2.3	Related Work	4
2.4	Summary	4
3	PROBLEM FORMULATION AND OBJECTIVES	5
3.1	Problem Statement	5
3.2	Objectives of the System	6
4	SYSTEM DESIGN	6
4.1	Introduction	6
4.2	System Architecture	6
4.3	Database Design	7
4.4	System Requirements	7
4.4.1	Hardware Requirements	7

SI NO	TITLE	PAGE NO
4.4.2	Software Requirements	7
5	SYSTEM IMPLEMENTATION	8
5.1	System Methodology	8
5.2	Module Descriptions	9
5.2.1	Landing Page / Gateway Module	10
5.2.2	User Registration Module	11
5.2.2.1	Customer Login Module	12
5.2.3	Service Status Tracking Module	13
5.2.4	Payment Module	15
5.2.5	Service Availability Module	17
5.2.6	Date and Time Slot Booking Module	19
6	RESULTS AND DISCUSSION	20
6.1	System Output Screens	20
6.2	Performance Evaluation	21
6.3	System Testing	22
7	CONCLUSION AND FUTURE WORK	22
7.1	Conclusion	22
7.2	Future Enhancements	23
	REFERENCES	24
	APPENDICES	25
	pseudocode	29

LIST OF ABBREVIATIONS

STPS – Smart Travel Planning System

DB – Database

SQL – Structured Query Language

API – Application Programming Interface

UI – User Interface

HTML – HyperText Markup Language

JS – JavaScript

HTTP – HyperText Transfer Protocol

SMTP – Simple Mail Transfer Protocol

CRUD – Create, Read, Update, Delete

CHAPTER 1

INTRODUCTION

1.1 General

The Online Vehicle Service System is an advanced web-based platform developed to automate and manage vehicle servicing operations efficiently. Vehicle maintenance is an essential requirement for ensuring vehicle safety, performance, and durability. In traditional service centers, most operations such as appointment booking, vehicle inspection tracking, billing, and customer communication are handled manually. This manual approach often results in inefficient management, increased paperwork, communication delays, booking conflicts, and poor customer satisfaction.

The Online Vehicle Service System addresses these problems by providing a centralized digital platform where customers can easily book service appointments, track service status, receive notifications, and access maintenance history. The system allows customers to interact with service centers remotely through a secure online interface, thereby reducing the need for physical visits during appointment scheduling and status verification.

The application helps service centers streamline daily operations by automating appointment scheduling, customer management, billing, spare parts inventory management, and service tracking. Service advisors and mechanics can update service progress in real time, enabling customers to receive accurate and timely updates regarding their vehicles.

1.2 Need for Online Vehicle Service Booking

With the rapid increase in the number of vehicles such as cars, bikes, and scooters, regular servicing and maintenance have become essential for vehicle safety and performance. Traditional vehicle service systems mainly depend on manual appointment booking, paperwork, and direct communication with service centers, which often leads to delays, inefficient management, and poor customer experience.

The Online Vehicle Service System is developed to automate and simplify vehicle servicing operations through a digital platform. The system allows customers to book service appointments online, track service status, receive notifications, and make secure online payments without visiting the service center physically.

The traditional servicing method has several limitations such as:

- Long waiting time
- Manual record maintenance
- Lack of real-time updates
- Poor customer communication
- Increased paperwork
- Higher possibility of human errors

The Online Vehicle Service System helps overcome these problems by providing:

- Online appointment booking
- Real-time service tracking
- Digital customer and vehicle records
- Secure payment processing
- Automated notifications and reminders

The system improves operational efficiency, reduces manual workload, enhances communication between customers and service centers, and increases overall customer satisfaction.

In conclusion, the Online Vehicle Service System is necessary to modernize vehicle service management and provide a faster, more reliable, and user-friendly solution for both customers and service centers.

1.3 Scope of the System

The scope of the Online Vehicle Service System is to provide an efficient and automated platform for managing vehicle servicing operations through an online environment. The system is designed to simplify the interaction between customers and vehicle service centers by digitizing appointment booking, service tracking, billing, and customer management processes.

The system allows customers to:

- Register and log in securely
- Add vehicle details
- Book vehicle service appointments online
- Select service types and preferred time slots
- Track service status in real time
- View booking and service history
- Make secure online payments
- Receive notifications and service reminders.

In conclusion, the scope of the Online Vehicle Service System covers complete digital management of vehicle servicing operations, providing a reliable, secure, and user-friendly solution for modern automobile service centers.

1.4 Organization of the Report

The Online Vehicle Service System project is organized into different chapters to explain the development and implementation of the system in a structured manner.

Chapter 1 – Introduction

This chapter provides an overview of the Online Vehicle Service System, including the need for the system, objectives, scope, and problems in the existing manual service process.

Chapter 2 – Literature Survey

This chapter discusses existing vehicle service management systems, related works, and the limitations of traditional servicing methods.

Chapter 3 – Existing and Proposed System

This chapter explains the drawbacks of the existing system and describes the proposed online vehicle service system along with its advantages and features.

Chapter 4 – System Design

This chapter includes system architecture, database design, data flow diagrams, use case diagrams, and hardware and software requirements.

Chapter 5 – System Methodology

This chapter explains the development process followed for the project, including requirement analysis, design, development, testing, deployment, and maintenance.

Chapter 6 – Module Description

This chapter describes the various modules such as:

Registration Module

Login Module

Vehicle Management Module

Service Booking Module

Payment Module

Notification Module

Admin Module

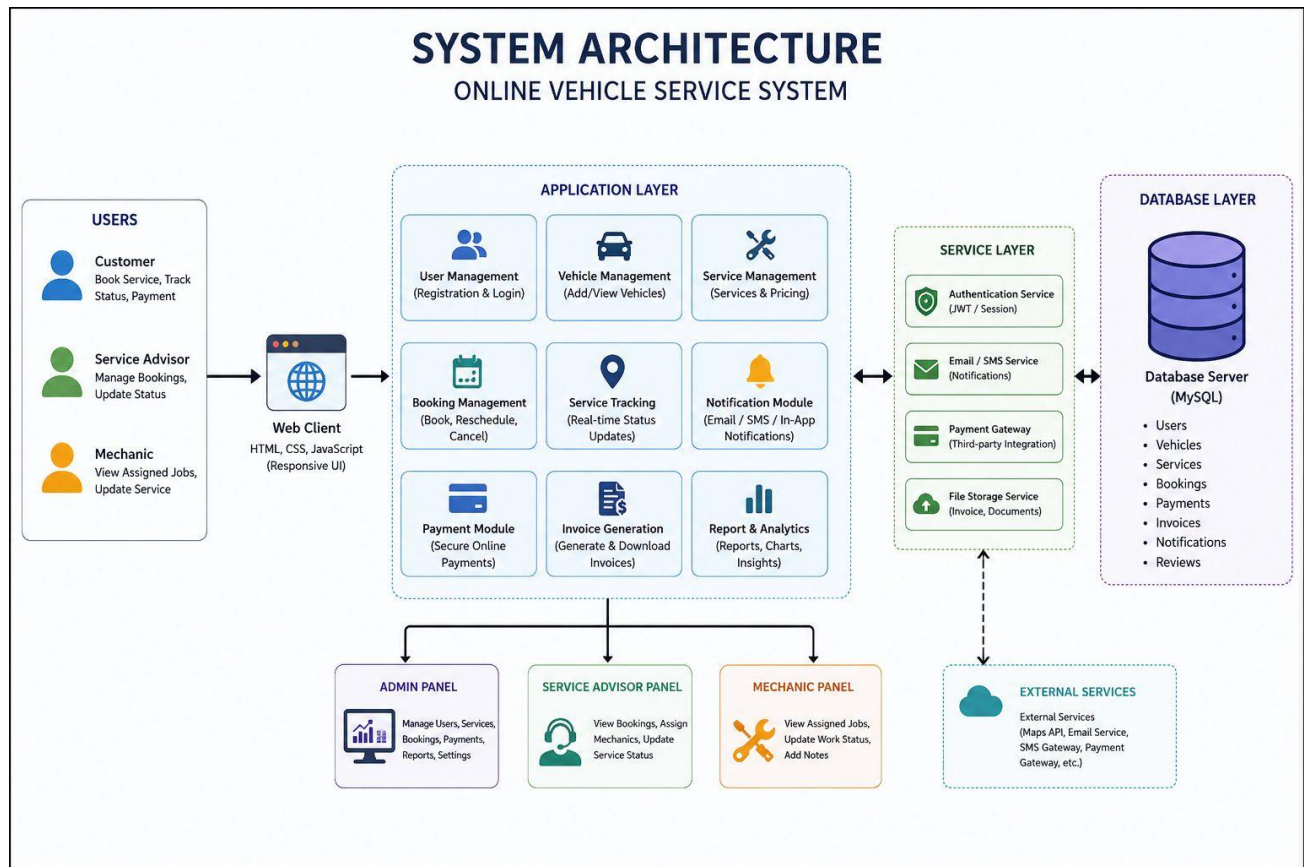
Chapter 7 – System Implementation and Results

This chapter explains the technologies used, source code logic, system implementation process, results obtained, and performance evaluation.

Chapter 8 – Conclusion and Future Work

This chapter summarizes the project and discusses future enhancements such as mobile application support, AI integration, and cloud-based management

System architecture diagram:



CHAPTER 2

LITERATURE SURVEY

2.1 Introduction

The literature survey is an important phase in project development that involves studying existing systems, technologies, research papers, and related applications connected to the Online Vehicle Service System. It helps in understanding current vehicle service management methods and identifying the limitations of traditional servicing systems.

Traditional vehicle service systems mainly depend on manual appointment booking, paperwork, and direct communication with service centers. These methods often cause delays, poor record management, booking conflicts, and lack of real-time service updates.

The literature survey studies various online vehicle service platforms and workshop management systems that provide features such as:

- Online appointment booking
- Service status tracking
- Digital billing

Vehicle maintenance history
Notification and reminder systems
Online payment facilities

The study of existing systems helps identify their advantages and limitations and provides ideas for developing a better and more efficient solution. The proposed Online Vehicle Service System aims to improve customer convenience, reduce manual work, and provide secure and reliable digital vehicle service management.

In conclusion, the literature survey provides a foundation for understanding the technologies and methods used in vehicle service systems and helps in designing an effective and user-friendly application

2.2 Existing Vehicle Service Booking

The existing online vehicle service booking systems are digital platforms developed to simplify vehicle servicing operations and improve customer convenience. These systems allow customers to book vehicle service appointments through websites or mobile applications instead of visiting service centers physically. Modern vehicle service platforms provide features such as online appointment scheduling, service tracking, digital billing, and customer management.

Most existing systems provide the following functionalities:

Customer registration and login
Online appointment booking
Service package selection
Vehicle information management
Real-time service status tracking
Automated notifications and reminders
Online payment processing
Booking cancellation and rescheduling
Digital invoice generation

Existing systems help reduce manual paperwork and improve workshop efficiency by automating booking and scheduling processes. Customers can select service types, choose preferred time slots, and receive confirmation messages instantly. Some advanced systems also provide real-time updates regarding service progress and maintenance schedules.

However, many existing systems still have certain limitations such as:

Limited real-time communication
Complex user interfaces
Lack of personalized service recommendations
Booking conflicts during peak hours
Limited integration with inventory management
Dependency on internet connectivity

Some systems also fail to provide complete transparency regarding service cost estimation and vehicle tracking.

The proposed Online Vehicle Service System is designed to overcome these limitations by providing a secure, user-friendly, and efficient platform with improved booking management, service tracking, automated notifications, and better customer interaction features.

2.3 Related Work

Several researchers and developers have worked on vehicle service management systems to improve workshop efficiency and customer convenience. Existing studies and applications mainly focus on automating appointment booking, service tracking, billing, and customer management processes.

A web-based Online Vehicle Service Management System was developed to overcome the limitations of manual servicing processes by providing online slot booking, task scheduling, and service management features. The system also included doorstep vehicle pickup and mechanic task allocation to improve customer convenience and workshop organization.

Another Vehicle Service Management System developed using PHP and MySQL provided separate modules for users and administrators. The system supported service requests, enquiry management, service status tracking, invoice generation, mechanic management, and customer record maintenance.

Research on online vehicle service center management systems also focused on improving operational efficiency through automated record keeping, report generation, and customer relationship management. These systems helped reduce paperwork and improved data organization in vehicle workshops.

Modern vehicle management platforms additionally provide features such as:

- Real-time vehicle tracking
- Automated notifications
- Inventory management
- Online payment systems
- Booking and scheduling management

Some advanced automotive platforms such as Openbay provide online automotive service marketplaces where customers can compare service providers, pricing, and customer ratings before booking appointments.

The study of these existing systems helped identify important functionalities and limitations. Based on the related work analysis, the proposed Online Vehicle Service System is designed to provide a more secure, user-friendly, efficient, and scalable platform with improved booking management, service tracking, notification services, and customer interaction features.

2.4 Summary

The literature survey provided a detailed study of existing vehicle service management systems, online booking platforms, and related technologies used in automobile service centers. The study showed that traditional vehicle servicing methods mainly depend on manual appointment booking, paperwork, and direct communication, which often lead to delays, inefficient management, and poor customer experience.

Several existing online vehicle service systems provide features such as:

- Online appointment booking
- Service status tracking
- Digital billing
- Vehicle history management
- Online payment processing
- Notification and reminder services

The related work analysis helped identify the advantages and limitations of current systems. Although many systems improve operational efficiency, some still face issues such as booking conflicts, limited real-time communication, complex interfaces, and lack of advanced automation features.

CHAPTER 3

PROBLEM FORMULATION AND OBJECTIVES

3.1 Problem Statement

In the current vehicle servicing process, most service centers still depend on manual methods for appointment booking, customer management, billing, and service tracking. Customers are often required to visit service centers physically or contact service advisors through phone calls to schedule appointments and obtain updates regarding vehicle servicing. This manual process consumes more time, increases paperwork, and reduces operational efficiency.

Traditional vehicle service systems face several problems such as:

- Long waiting time for appointment scheduling
- Manual record maintenance
- Poor communication between customers and service centers
- Difficulty in tracking service status
- Booking conflicts and scheduling errors
- Lack of real-time updates

Increased chances of human errors

Inefficient billing and payment management

Customers also face inconvenience in checking service availability, receiving maintenance reminders, and accessing vehicle service history. Service centers find it difficult to manage customer records, appointments, spare parts inventory, and billing operations efficiently using manual methods.

To overcome these problems, there is a need for an automated Online Vehicle Service System that provides online appointment booking, real-time service tracking, secure payment processing, notification services, and efficient workshop management through a centralized digital platform.

The proposed system aims to simplify vehicle servicing operations, improve customer convenience, reduce manual workload, enhance communication, and provide better management of vehicle maintenance activities.

3.2 Objectives of the System

The main objective of the Online Vehicle Service System is to automate and simplify vehicle servicing operations through a secure and user-friendly online platform. The system is designed to improve customer convenience, reduce manual workload, and enhance the overall efficiency of vehicle service management.

The major objectives of the system are:

- To provide online vehicle service appointment booking
- To reduce manual paperwork and administrative work
- To maintain customer and vehicle records digitally
- To provide real-time service status tracking
- To improve communication between customers and service centers
- To enable secure online payment processing
- To prevent booking conflicts and scheduling errors
- To maintain vehicle service history efficiently
- To send automated notifications and service reminders
- To improve workshop management and operational efficiency
- To enhance customer satisfaction through faster and reliable services
- To provide a centralized platform for managing vehicle servicing activities

The system also aims to provide a scalable and reliable solution that can be further enhanced with advanced technologies such as mobile applications, AI-based maintenance prediction, and cloud-based management systems in the future.

CHAPTER 4

SYSTEM DESIGN

4.1 Introduction

System design is an important phase in software development that defines the overall structure, architecture, modules, database design, and workflow of the application. It acts as a bridge between requirement analysis and system implementation. The main purpose of system design is to convert user requirements into a structured and efficient solution that can be easily developed, maintained, and upgraded.

In the Online Vehicle Service System, the system design focuses on developing a secure, user-friendly, and efficient platform for managing vehicle servicing operations digitally. The design helps organize different functionalities such as customer registration, login authentication, vehicle management, service booking, service tracking, payment processing, notifications, and administrative control.

The system is designed using a modular architecture where each module performs a specific function while interacting with other modules smoothly. The application follows a three-tier architecture consisting of:

Presentation Layer
Application Layer
Database Layer

4.2 System Architecture

The system architecture of the Online Vehicle Service System represents the overall structure and interaction between different components of the application. It defines how users, servers, modules, and databases communicate to perform vehicle service management operations efficiently.

- **Presentation Layer:**

This layer represents the user interface where users and administrators interact with the system. It includes web pages such as login pages, dashboards, travel planning pages, and itinerary views developed using HTML, CSS, and JavaScript.

- **Application Layer:**

This layer handles the business logic of the system. It processes user requests, verifies login credentials, manages travel plans, generates itineraries, and communicates with the database. The backend of the system is implemented using Node.js.

- **Database Layer:**

This layer stores all the system data including user details, travel destinations, itineraries, bookings, and travel history. The database ensures secure storage and easy retrieval of information when required.

This architecture ensures smooth communication between users and the database while maintaining system efficiency and reliability.

4.3 System Requirements

System requirements define the hardware and software resources needed to run the application successfully.

4.3.1 Hardware Requirements

The Smart Travel Planning System does not require high-end hardware and can run on standard computers.

- Processor: Intel Core i3 or higher
- RAM: Minimum 4 GB
- Storage: 500 GB Hard Disk or SSD
- Input Devices: Keyboard and Mouse
- Internet Connectio

4.3.2 Software Requirements

The system is developed using commonly available software tools and technologies

- Operating System: Windows 10 or later
- Frontend Technologies: HTML, CSS, JavaScript
- Backend Technology: Node.js
- Database: MySQL
- Development Environment: Visual Studio Code
- WebBrowser: Google chrome

CHAPTER 5

SYSTEM IMPLEMENTATION

5.1 System Methodology

The system methodology describes the approach and process followed for the development and implementation of the Online Vehicle Service System. It explains the various stages involved in designing, developing, testing, and deploying the application successfully.

The Online Vehicle Service System follows a structured software development methodology to ensure proper planning, efficient implementation, and reliable system performance. The methodology helps in organizing the development process systematically and ensures that the project requirements are achieved effectively.

The project follows the Waterfall Model methodology because it provides a simple and step-by-step approach for software development. In this model, each phase is completed before moving to the next phase.

5.2 Module Descriptions

The Online Vehicle Service System is divided into different modules to simplify system operations and improve efficiency. Each module performs a specific function and works together to provide complete vehicle service management.

1. Landing / Gateway Module

Provides access to the website and navigation to different pages such as login, registration, services, and contact information.

2. Customer Registration Module

Allows new users to create accounts by entering personal and vehicle details securely.

3. Customer Login Module

Provides secure authentication for registered users using username and password.

4. Homepage Module

Displays service information, offers, service categories, and website navigation options.

5. Vehicle Management Module

Manages customer vehicle details and maintains vehicle service history.

6. Service Booking Module

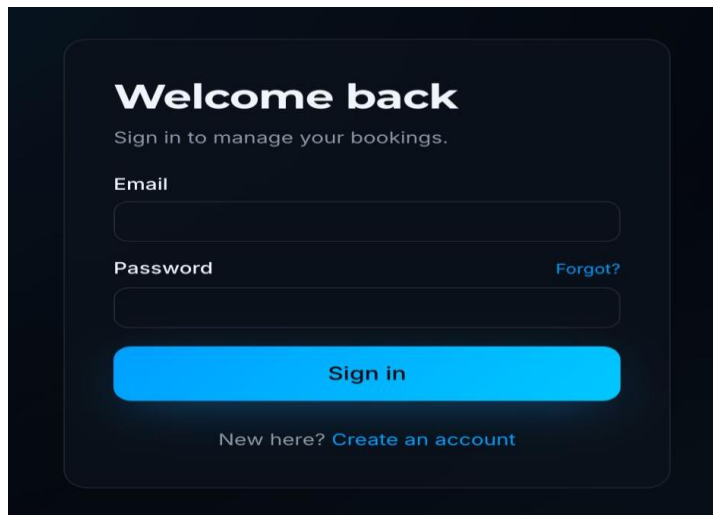
Allows customers to book service appointments by selecting service type, date, and time.

7. Service Status Tracking Module

Helps customers track the real-time status of vehicle servicing operations.

8. Payment Module

Handles secure online payment processing and invoice generation.



5.3 Landing Page / Gateway Module

The Landing Page or Gateway Module is the main entry point of the Online Vehicle Service System. It provides the first interaction between users and the application and acts as a bridge between customers and the various services offered by the system. The module is designed with a simple, attractive, and user-friendly interface to help users navigate the website easily.

The primary objective of the Landing Page / Gateway Module is to provide quick access to important features such as registration, login, service booking, vehicle management, and contact information. It also displays information about available vehicle services, special offers, and workshop details to improve customer engagement.

Features of the Landing Page / Gateway Module

- Displays the homepage of the application
- Provides navigation menus for easy browsing
- Displays available vehicle service categories
- Shows promotional banners and special offers
- Provides login and registration access
- Displays contact details and customer support information
- Redirects users to booking and vehicle management modules
- Provides quick access to appointment booking services

Working of the Module

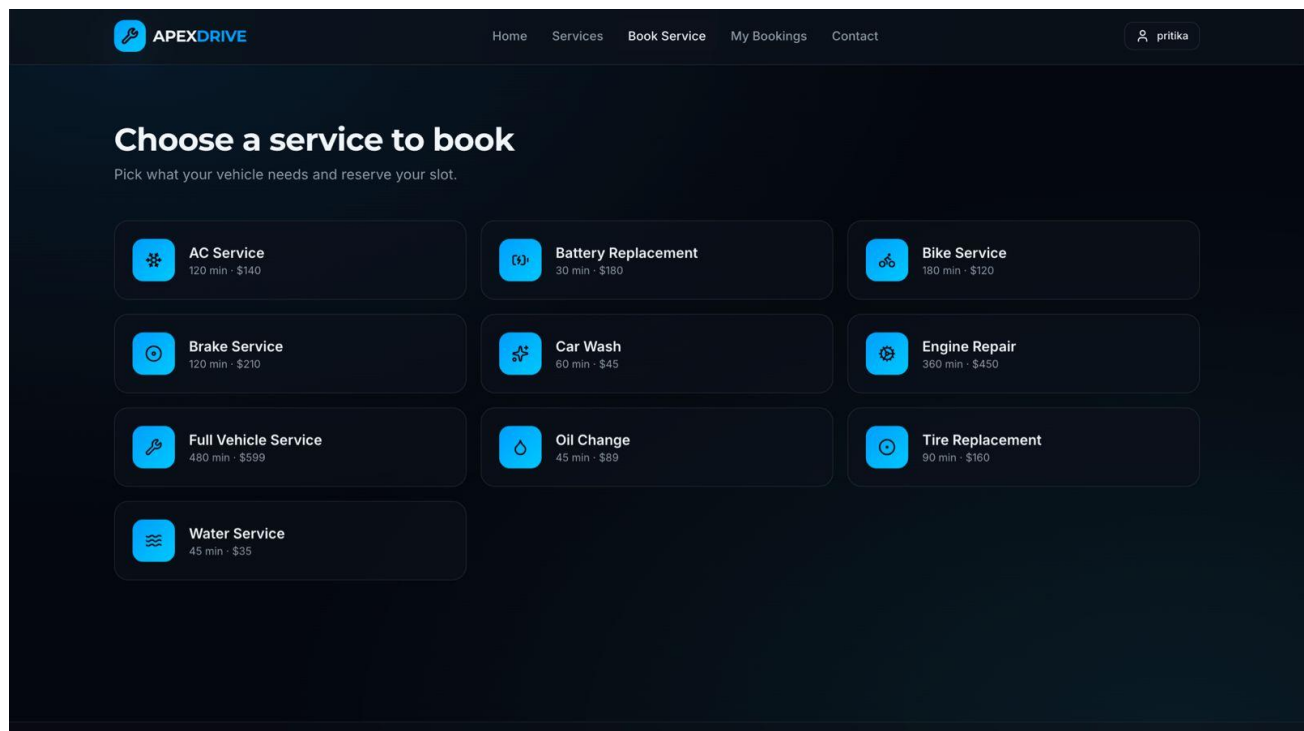
- The user opens the Online Vehicle Service System website.
- The homepage is displayed with navigation menus and service information.
- Users can browse available services and workshop details.
- Customers can register or log in to access booking functionalities.
- The system redirects users to the selected module such as service booking or vehicle management.

Advantages of the Landing Page / Gateway Module

- Provides easy navigation throughout the website
- Improves customer interaction and accessibility

Enhances user experience with a simple interface
Allows quick access to important services
Helps attract customers through offers and service information
Conclusion

The Landing Page / Gateway Module plays an important role in the Online Vehicle Service System by providing an attractive and user-friendly interface for customers. It simplifies navigation, improves accessibility to services, and acts as the central access point for all system functionalities.



5.3.1 User Registration Module

The User Registration Module is an important component of the Online Vehicle Service System. This module allows new customers to create accounts in the system by entering their personal and vehicle-related information. Registration helps the system identify users and provide personalized vehicle service management features.

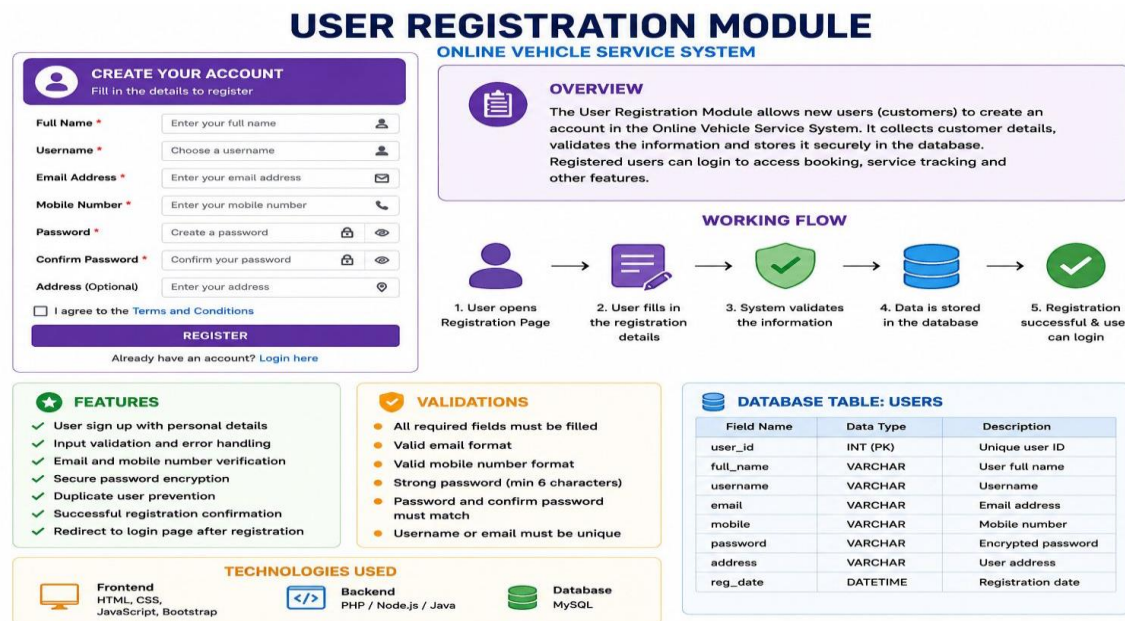
The main objective of this module is to securely collect and store customer details so that users can access online service booking, vehicle management, payment processing, and service tracking functionalities.

Features of the User Registration Module
New customer account creation

Input validation for accurate data entry
 Secure password storage
 Duplicate account prevention
 Vehicle information registration
 Database integration for record storage
 Registration Details

Customers are required to enter details such as:

Full Name
 Email Address
 Mobile Number
 Address
 Username
 Password
 Vehicle Number
 Vehicle Model



5.3.2 Customer Login Module

The Customer Login Module is an important component of the Online Vehicle Service System that provides secure authentication for registered users. This module allows customers to access their accounts using valid login credentials such as username/email and password. It ensures that only authorized users can access booking, vehicle management, payment, and service tracking functionalities.

The main objective of the Customer Login Module is to provide secure and reliable access to the system while protecting customer information from unauthorized access.

Features of the Customer Login Module

Secure user authentication

Username and password validation

Session management

Forgot password functionality

Error handling for invalid credentials

Secure access to customer dashboard

Working of the Customer Login Module

The customer opens the login page.

The user enters username/email and password.

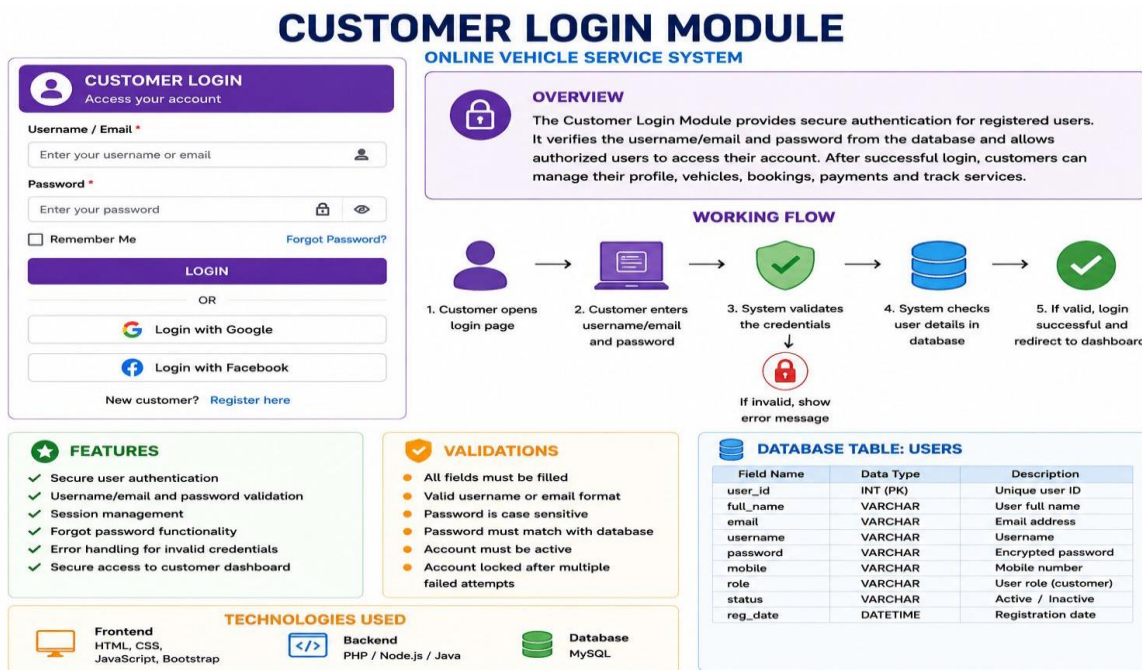
The system validates the entered credentials.

The application checks user details from the database.

If the credentials are correct, the user is successfully logged in.

The customer is redirected to the dashboard for accessing booking and service management features.

If the credentials are incorrect, an error message is displayed.



5.2.3 Service Status Tracking Module

The Service Status Tracking Module is an important component of the Online Vehicle Service System that allows customers to monitor the progress of their vehicle servicing in real time. This module improves transparency and communication between customers and service centers by providing regular updates about the current status of vehicle maintenance activities.

The main objective of this module is to help customers track their vehicle service progress without visiting the service center physically or contacting service advisors repeatedly.

Features of the Service Status Tracking Module

Real-time service progress tracking

Service stage updates

Service completion notifications

Vehicle delivery status monitoring

Customer service transparency

Automated status alerts

Service Status Stages

The module displays different stages of vehicle servicing such as:

Service Request Received

Vehicle Inspection

Service In Progress

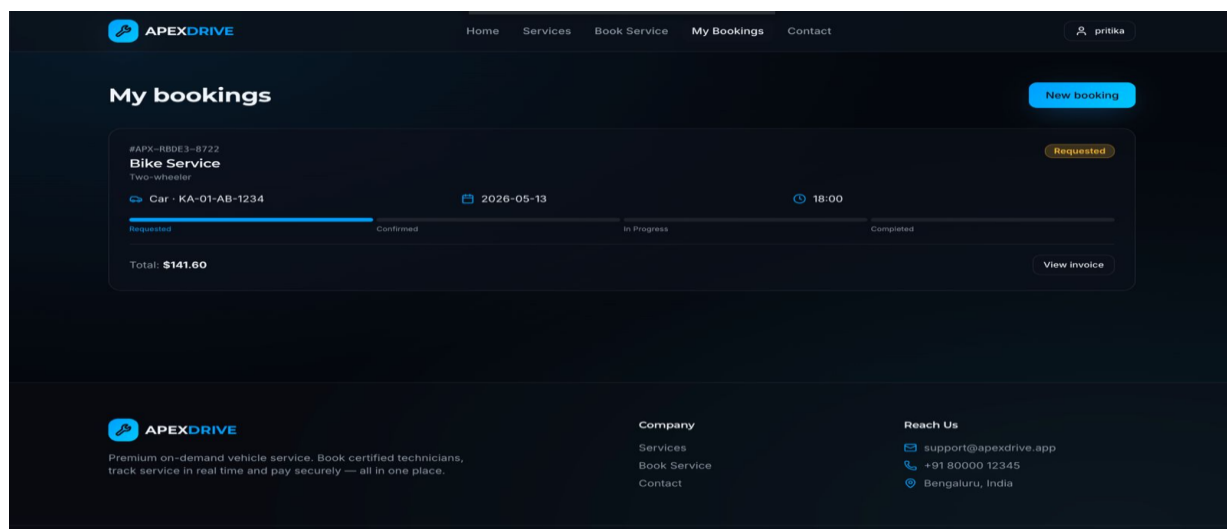
Spare Parts Replacement

Quality Checking

Service Completed

Ready for Delivery

Customers can view the current status of their vehicle directly through the application.



5.2.3 Payment Module

The Payment Module is an essential component of the Online Vehicle Service System that handles secure online payment processing for vehicle servicing operations. This module allows customers to make payments digitally after booking vehicle services, improving convenience and reducing manual billing processes.

The main objective of the Payment Module is to provide a secure, reliable, and efficient payment system for customers and service centers.

Features of the Payment Module

Secure online payment processing

Multiple payment methods support

Automatic invoice generation

Payment confirmation notifications

Transaction validation

Billing and payment history management

Payment Methods Supported

The system supports different online payment methods such as:

Credit Card

Debit Card

UPI Payments

Net Banking

Digital Wallets

This provides flexibility and convenience for customers.

Working of the Payment Module

The customer books a vehicle service appointment.

The system generates the service bill.

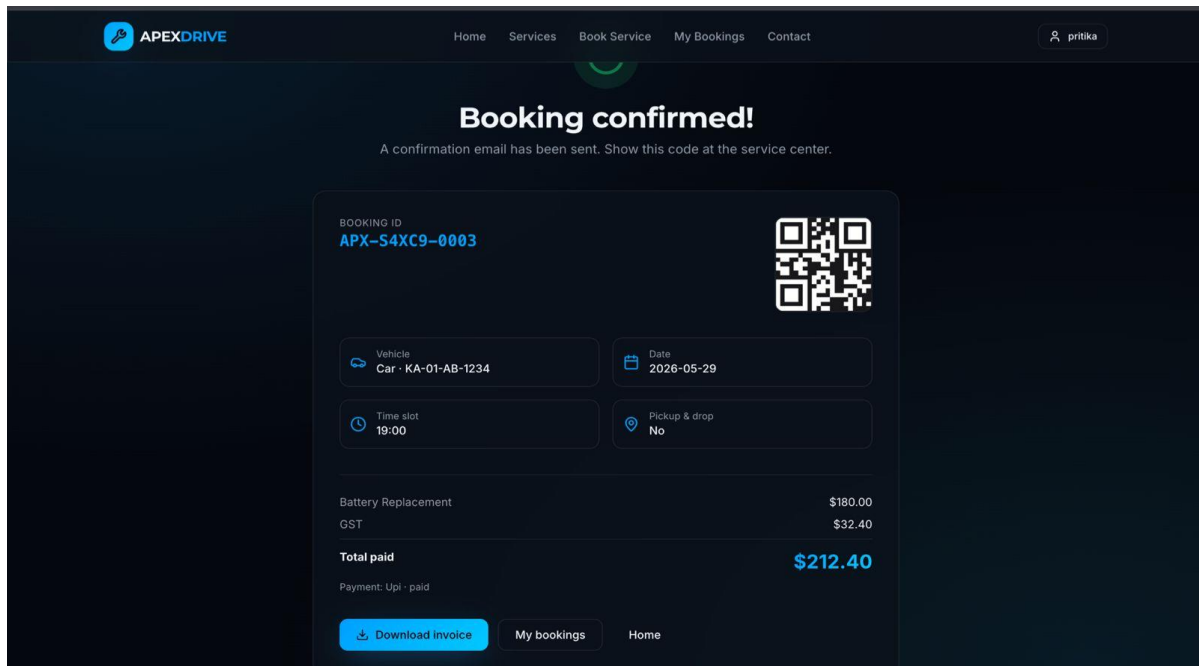
The customer selects a preferred payment method.

Payment details are securely processed through the payment gateway.

The system verifies the transaction.

Payment confirmation is displayed to the customer.

Invoice and payment records are stored in the database.



5.2.5 Service Availability Module

The Service Availability Module is an important part of the Online Vehicle Service System that allows customers to view available vehicle services, service slots, and appointment timings in real time. This module helps customers easily select suitable services and available schedules based on their convenience.

The main objective of this module is to manage service availability efficiently and prevent booking conflicts by displaying real-time service information.

Features of the Service Availability Module

Displays available vehicle services

Shows available appointment slots

Real-time availability updates

Service timing management

Slot booking validation

Prevents duplicate bookings

Services Displayed

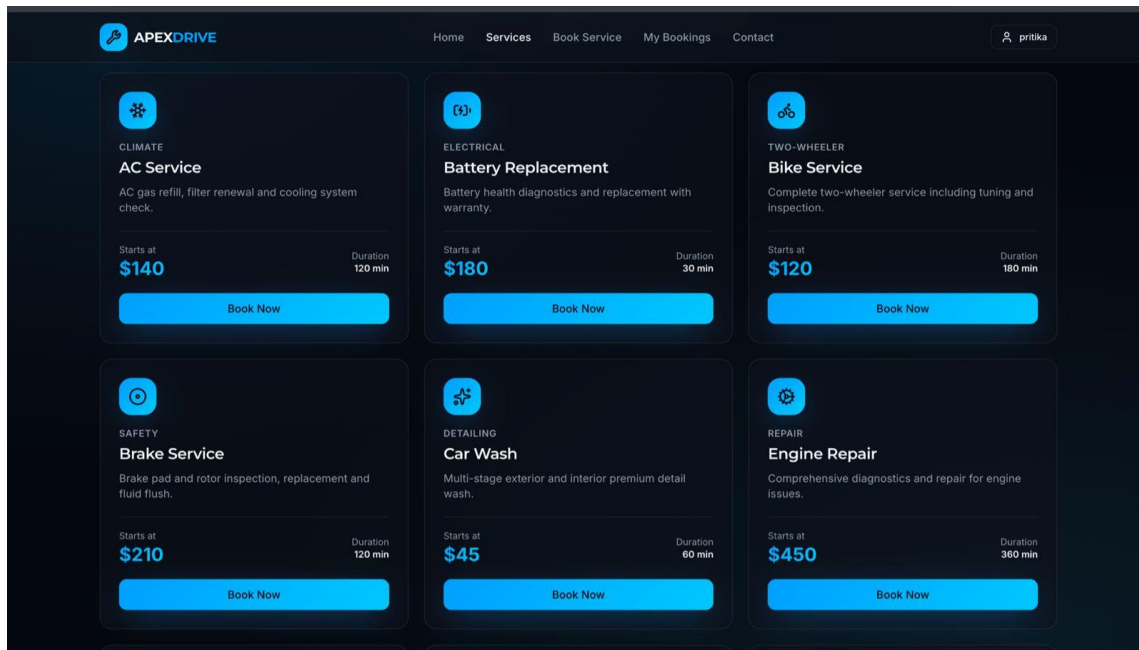
The module displays different vehicle services such as:

Oil Change

AC Service

Brake Service
Car Wash
Engine Repair
Battery Replacement
Tire Replacement
Full Vehicle Service

Customers can view service duration, pricing, and slot availability before booking.



CHAPTER 6

RESULTS AND DISCUSSION

6.1 Results

The Online Vehicle Service System was successfully developed and implemented to automate vehicle servicing operations and improve customer convenience. The system was tested with different user activities such as registration, login, service booking, payment processing, and service tracking.

The results showed that the system performed efficiently and provided accurate management of vehicle service operations.

Observed Results:

- Successful customer registration and login
- Efficient online service booking
- Real-time service status tracking
- Secure payment processing
- Accurate booking and appointment management

Improved communication between customers and service centers

The system reduced manual workload and improved overall operational efficiency..

6.2 Performance Evaluation

- Performance evaluation was conducted to analyze the efficiency, reliability, and responsiveness of the Online Vehicle Service System under different conditions.
-
- Parameters Evaluated:
- System response time
- Booking accuracy
- Database performance
- Security and authentication
- User interaction efficiency
- Evaluation Results:
- Fast response during booking and login operations
- Accurate appointment scheduling without conflicts
- Secure authentication and payment processing
- Stable database connectivity and data retrieval
- Smooth user interface and navigation
-
- The system performed reliably with minimal errors and provided efficient service management.

6.3 Resources Utilization

Resource utilization analysis was performed to evaluate how effectively the system use hardware and software resources during operation.

Resources Utilized:

CPU usage

Memory usage

Database storage

Network bandwidth

Observations:

Efficient memory utilization during user operations

Low CPU consumption for regular service requests

Optimized database storage management

Stable network performance during online booking and payment processing

The system utilized resources efficiently and maintained stable performance during testing..

CHAPTER 7

CONCLUSION AND FUTURE WORK

7.1 Conclusion

The Online Vehicle Service System successfully automated vehicle servicing operations by providing an efficient, secure, and user-friendly platform for appointment booking, service tracking, payment processing, and workshop management.

The system reduced manual workload, improved customer communication, enhanced booking efficiency, and provided better operational control for service centers

7.2 Future Work

Future enhancements may include:

Mobile application support

AI-based maintenance prediction

GPS-based service center search

Cloud-based database management

QR code vehicle tracking

Multi-language support

Online spare parts ordering

These improvements can further enhance system intelligence and customer convenience.

REFERENCES

[1] Research on Online Vehicle Service Booking Platforms, 2022.

[2] Studies on AI-based Recommendation Systems in Vehicle Service Applications.

[3] Node.js Documentation – <https://nodejs.org>

[4] MySQL Documentation – <https://dev.mysql.com/doc>

[5] W3C – HTML Standards – <https://www.w3.org>

[6] D. Flanagan, JavaScript: The Definitive Guide, O'Reilly Media, 2020.

APPENDICES

Appendix-1: source code pseudocode

The pseudocode of the Online Vehicle Service Booking System represents the logical flow of the application. It explains the major functionalities and operations performed by the system in a simplified programming format. The pseudocode helps developers understand the working process before actual implementation.

System Initialization Logic

START

LOAD Configuration Files

CONNECT Database

LOAD Modules

CHECK Server Status

INITIALIZE Sessions

DISPLAY Homepage

STOP

User Login Logic

START

DISPLAY Login Page

INPUT Username or Email

INPUT Password

IF Username or Password is Empty THEN

 DISPLAY "Please Enter All Fields"

 STOP

END IF

CONNECT Database

SEARCH User Details using Username or Email

IF User Exists THEN

 VERIFY Password

 IF Password is Correct THEN

 CREATE User Session

 DISPLAY "Login Successful"

 REDIRECT to User Dashboard

 ELSE

 DISPLAY "Invalid Password"

 END IF

ELSE

 DISPLAY "User Does Not Exist"

END IF

CLOSE Database Connection

STOP

Customer Login

START

DISPLAY Login Form

INPUT Username

INPUT Password

VERIFY Credentials

```
IF Valid THEN
    OPEN Dashboard
ELSE
    DISPLAY "Invalid Credentials"
END IF

STOP
```

Service Booking

```
START

SELECT Vehicle
SELECT Service Type
SELECT Date and Time

CHECK Availability

IF Slot Available THEN
    CONFIRM Booking
    STORE Appointment Details
ELSE
    DISPLAY "Slot Not Available"
END IF

STOP
```

System Features Implemented

Implemented Features:

- Customer registration
- Secure login authentication
- Vehicle management
- Online service booking
- Real-time status tracking
- Spare parts management
- Online payment processing
- Notification alerts
- Admin dashboard

Report generation
FINAL SYSTEM STATUS

The Online Vehicle Service System was successfully developed and tested.

Appendix 2: System Test Verify

Test Case 1: Service Booking Creation

Result: PASS – plan created successfully

Test Case 2: Duplicate Detection

Result: PASS – duplicate entry prevented

Test Case 3: Service Plan Update

Result: PASS – plan updated correctly

Test Case 4: OTP Verification

Result: PASS – OTP sent and verified successfully

Test Case 5: Vehicle service Viewing

Result: PASS – users can view travel detail

Final System Status

Component	Status
Registration Module	Completed
Login Module	Completed
Vehicle Management	Completed
Service Booking	Completed
Service Tracking	Completed
Payment Module	Completed
Notification System	Completed
Database Connectivity	Active
Admin Dashboard	Completed

The system is fully functional and ready for deployment in real-world vehicle service centers.



HTML, CSS, and JavaScript: Building the Web

Course completed by PRITIKA R AI&DS
May 12, 2026 at 06:21AM UTC • 3 hours 37 minutes

Top skills covered

Web Development

Cascading Style Sheets (CSS)

HTML

Shea Hanson, Head of Learning Content Strategy

Certificate ID: b23e3782b653c526737887a41ed4218054d2770caca18fd774bb0173709dd707



CERTIFICATE OF COMPLETION

Pritika R

has successfully completed the online course:
Java Servlet Basics and JSP 101

This professional has demonstrated initiative and a commitment to deepening their skills and advancing their career. Well done!

12th May 2026
Certificate code : 10218723



Krishna Kumar
CEO, Simplilearn

